1. **Double Transposition Cipher:** Use two keywords to do a tabular transposition cipher twice.

   Encryption Steps

   i. Write the plaintext in **rows** using Keyword 1.

   ii. Rewrite columns with Keyword 1 in alphabetical order.

   iii. Read off the **columns** to form the new "plaintext".

   iv. Repeat this process with Keyword 2.

2. **Example:** Encrypt the word **TRANSPOSITION** using double transposition with

   Keyword 1: `KEY` and Keyword 2: `SHORT`.

3. How is this different from the tabular transposition with keyword encryption method we used earlier?

4. **Decryption:**
   i. Make four grid headers: Keyword 2 (alphabetical order), Keyword 2 (natural order), Keyword 1 (alphabetical order), Keyword 1 (natural order).
   ii. Count the number of characters in the ciphertext and determine grid dimensions.
   iii. Starting with Keyword 2 (alphabetical) grid, fill **columns**.
   iv. Copy columns into the grid for Keyword 2 (natural order).
   v. Read off the **rows** to find new ciphertext.
   vi. Repeat (iii) with Keyword 1 (alphabetical).
5. Decrypt the ciphertext from problem 2. (We know it should decrypt as TRANSPOSITION!)

6. Decrypt the ciphertext E T E T S P I H X C R assuming it was encrypted using the scheme above.